

Internship Report on " AUTO PARTS HUB"

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Executive Summary

This project report presents the development of **AutoParts Hub**, a full-stack web application designed to simplify the purchasing of automobile parts. The system allows users to register, login, browse products, add items to a cart, and place orders.

The backend is developed using **Java HttpServer**, and the database is managed using **Oracle with JDBC connectivity**. The frontend is designed using **HTML, CSS, and JavaScript**, ensuring an interactive and user-friendly experience.

This project demonstrates real-world concepts such as client-server communication, database management, and API integration.

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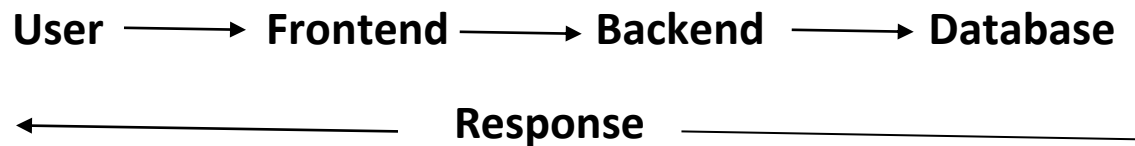
Preface

This project was developed as part of practical learning to understand full-stack web development. It helped in applying theoretical knowledge into real-world implementation.

The project includes:

- User authentication system
- Database storage
- Cart functionality
- Order system

This experience improved my problem-solving, debugging, and development skills.



INTRODUCTION

The AutoParts Hub system is an online platform designed to provide users with easy access to automobile parts.

Key Features:

User Registration and Login

Product Display

Add to Cart

Order Placement

Technologies Used:

Frontend: HTML, CSS, JavaScript

Backend: Java (HttpServer)

Database: Oracle DB

Connectivity: JDBC

OBJECTIVES

The main objectives of this project are:-

To develop a full-stack web application

To implement secure user authentication

To connect Java backend with Oracle database

To simulate an online shopping experience

PROBLEM STATEMENT

Traditional automobile parts purchasing systems face issues such as:

- Limited availability
- Time-consuming process
- Lack of centralized platform

Solution:

Develop a web-based platform that allows users to:

- Access products online
- Place orders easily
- Store data securely

EXISTING VS PROPOSED SYSTEM

Existing System:

Offline purchasing

Manual record keeping

Limited accessibility

Proposed System:

Online platform

Automated data storage

Fast and efficient system

SYSTEM DESIGN

◆ SYSTEM FLOWCHART

START



User Opens Website



Register / Login



[New User? → Register → Save in DB]



Login Validation



[Valid? → YES → Products Page | NO → Error]



Browse Products



Add to Cart



Place Order



Store Order



END

IMPLEMENTATION

◆ Backend Implementation

Developed using Java HttpServer

APIs created:

/register

/login

/order

◆ Database Implementation

USERS Table:

username

password

ORDERS Table:

order_id

username

total_amount

Frontend Implementation

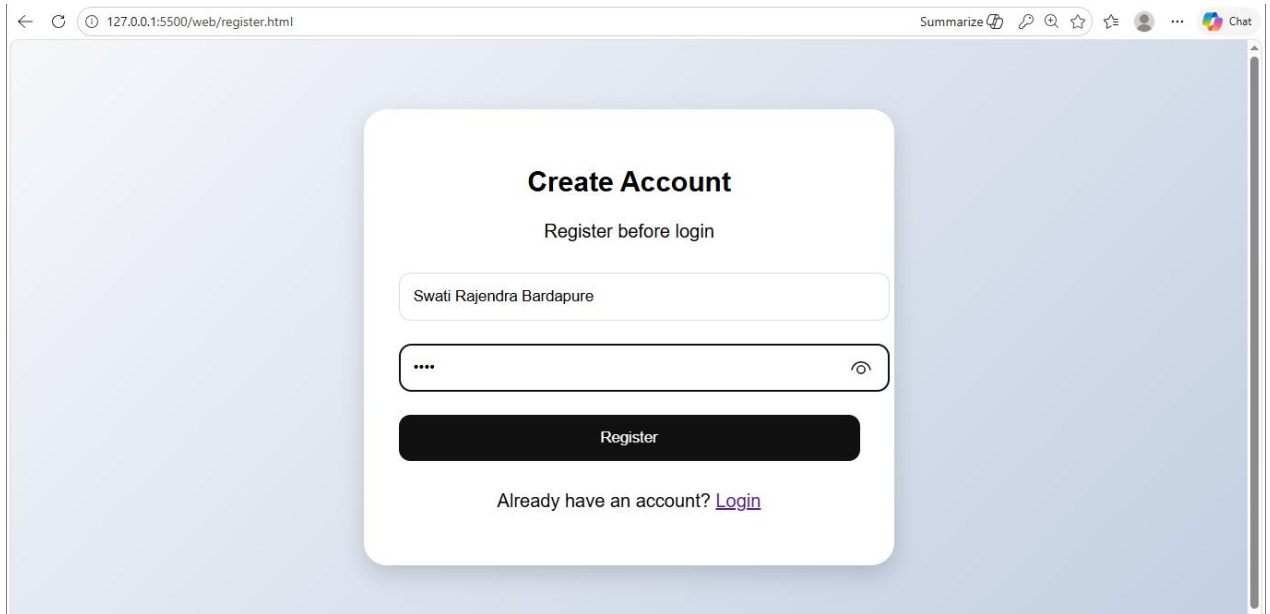
HTML forms for login/register

JavaScript Fetch API for backend communication

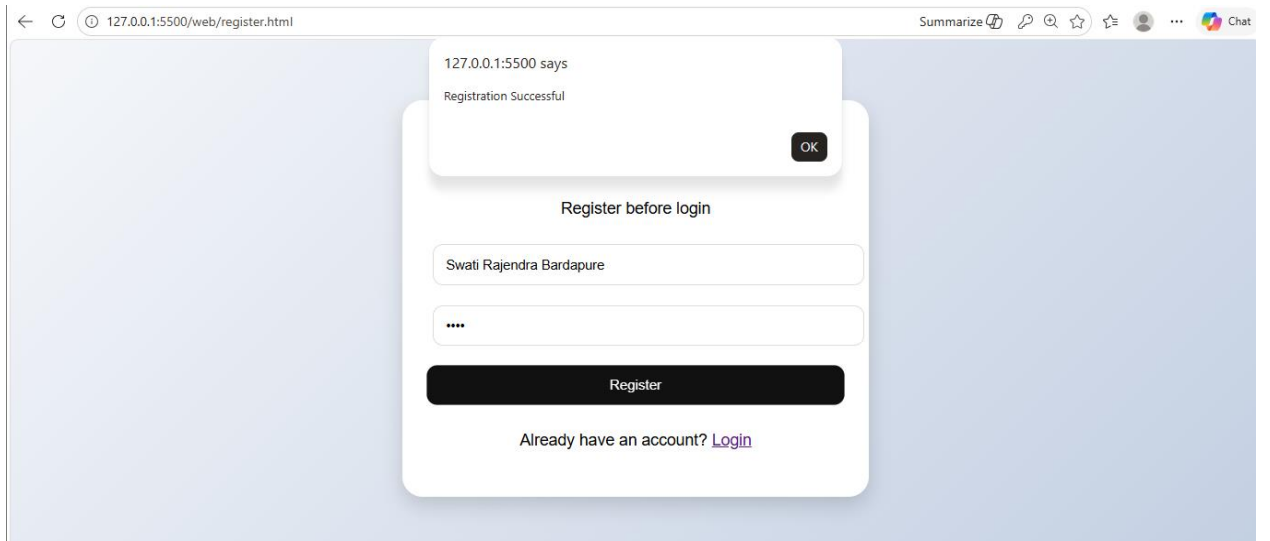
LocalStorage for cart

SYSTEM SCREENSHOTS

1. Registration Page



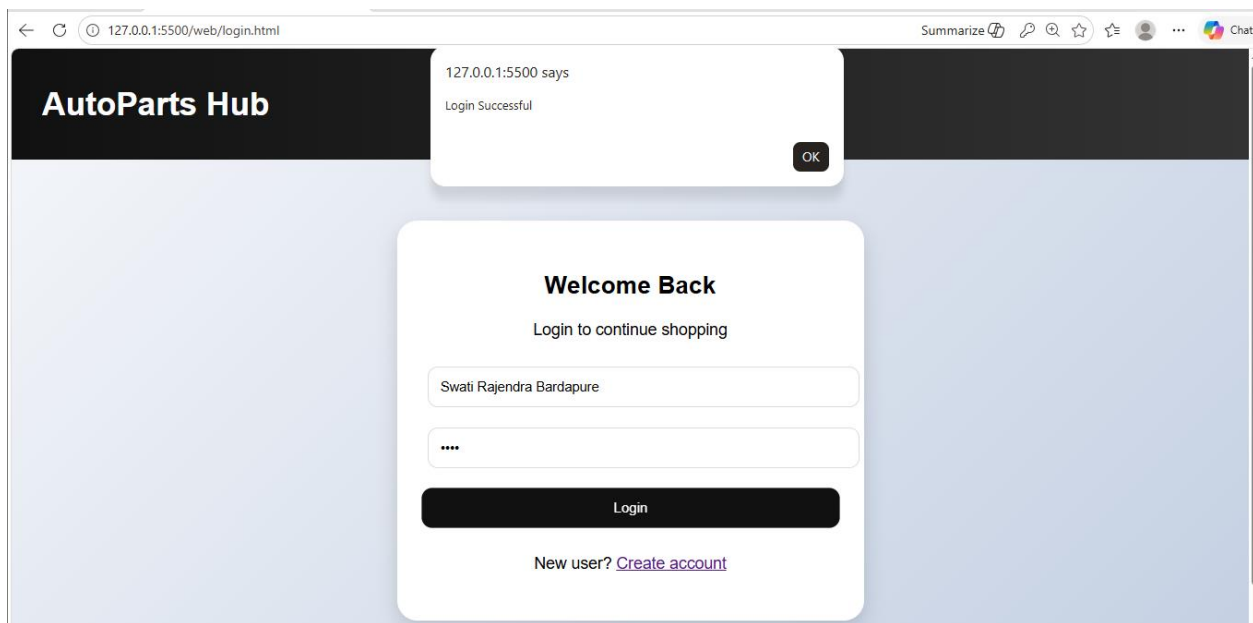
A screenshot of a web browser displaying the registration page. The browser's address bar shows the URL "127.0.0.1:5500/web/register.html". The page features a central white card with the heading "Create Account" and the subtext "Register before login". Below this, there are two input fields: the first contains the text "Swati Rajendra Bardapure", and the second is a password field with four dots and a toggle icon. A black "Register" button is positioned below the input fields. At the bottom of the card, it says "Already have an account? [Login](#)". The browser's top bar includes a "Summarize" button and a "Chat" icon.



A screenshot of the same registration page, but with a success message overlay. A white notification box at the top left of the card area displays the text "127.0.0.1:5500 says Registration Successful" and an "OK" button. The registration form below remains visible and unchanged, showing the same inputs and buttons as the previous screenshot.

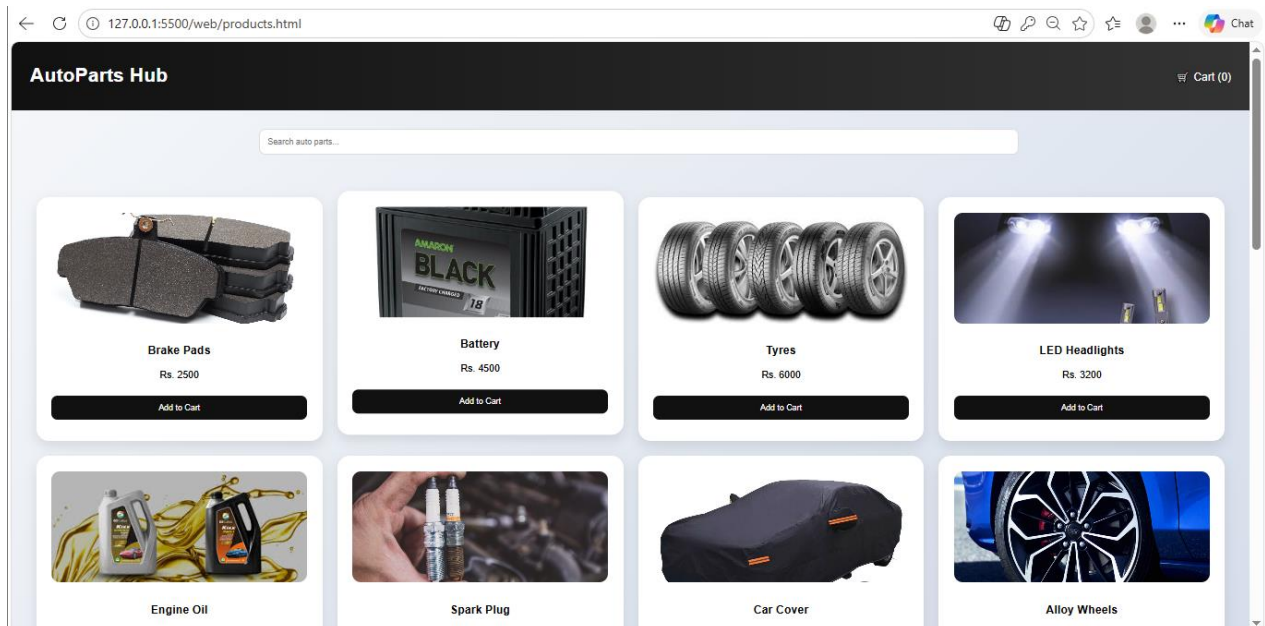
Caption: User Registration Interface

2. Login Page



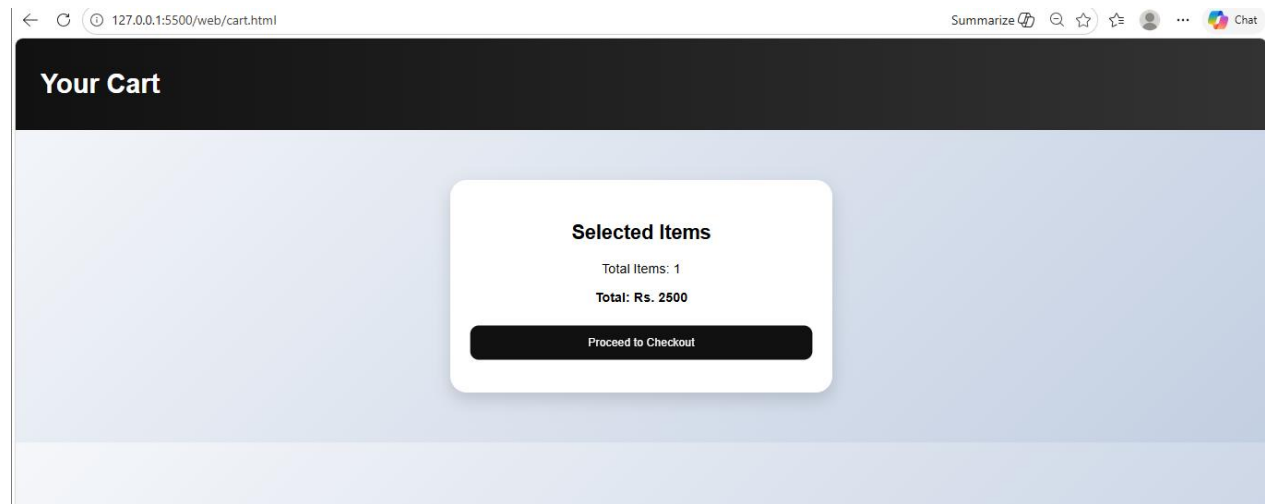
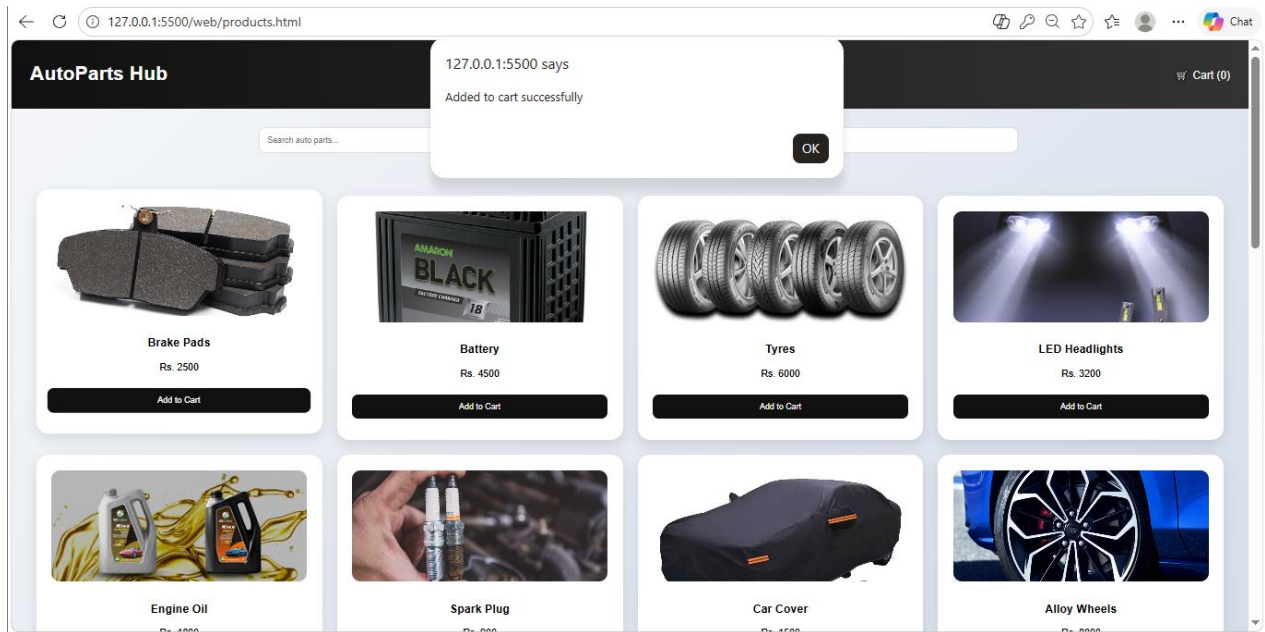
Caption: User Login Interface

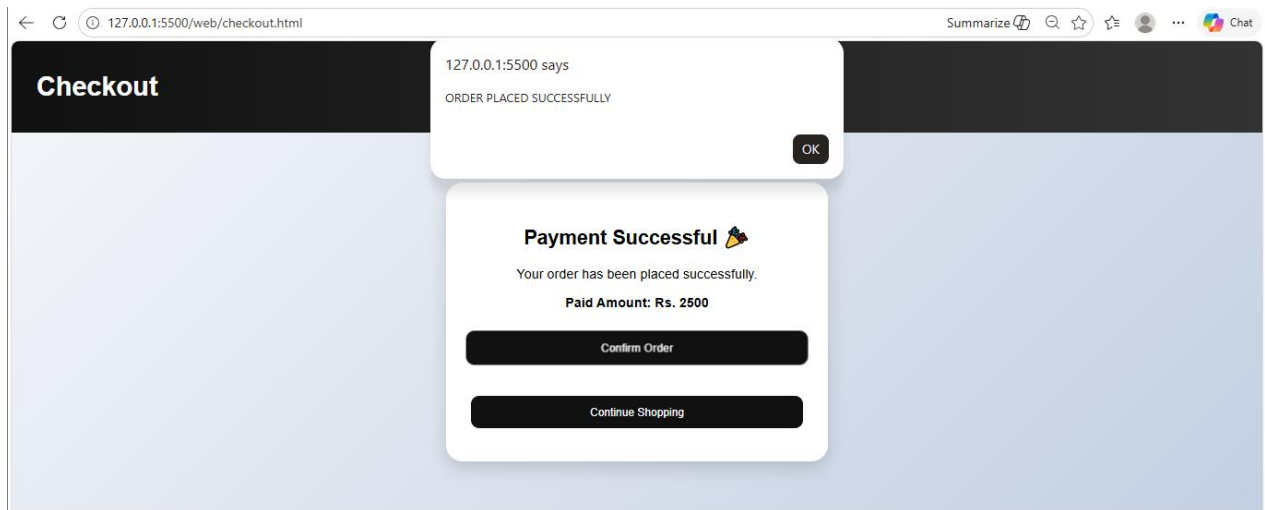
3. Products Page



Caption: Product Display

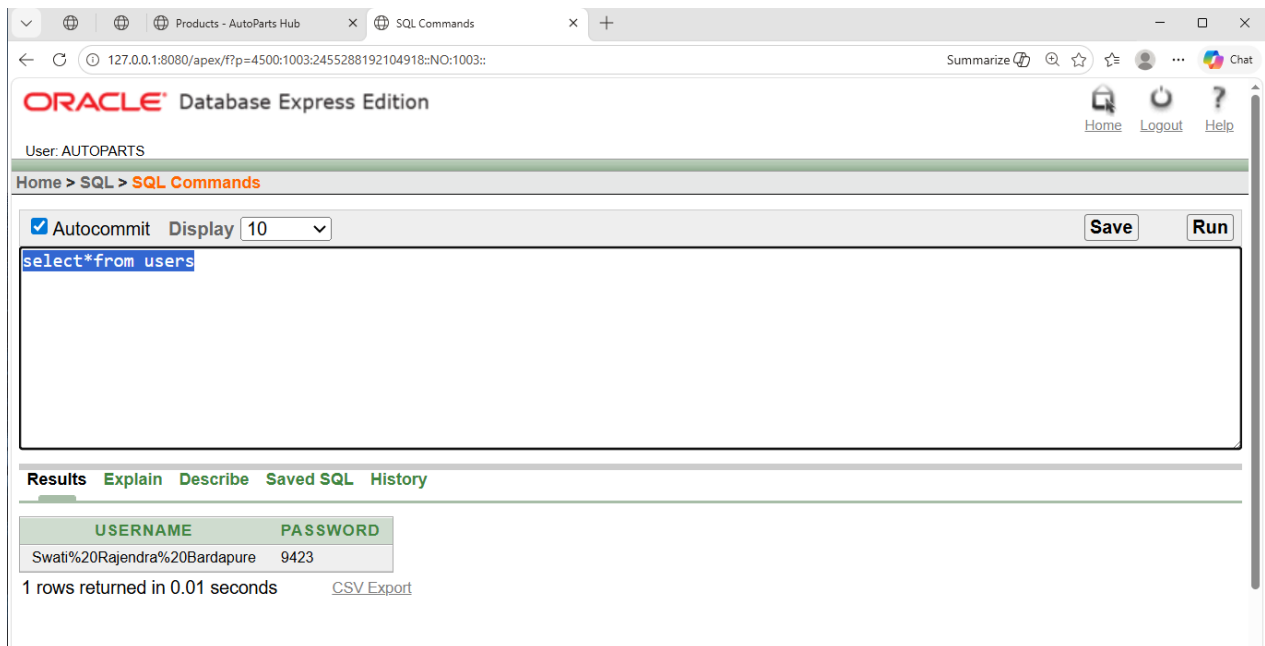
4. Cart Function



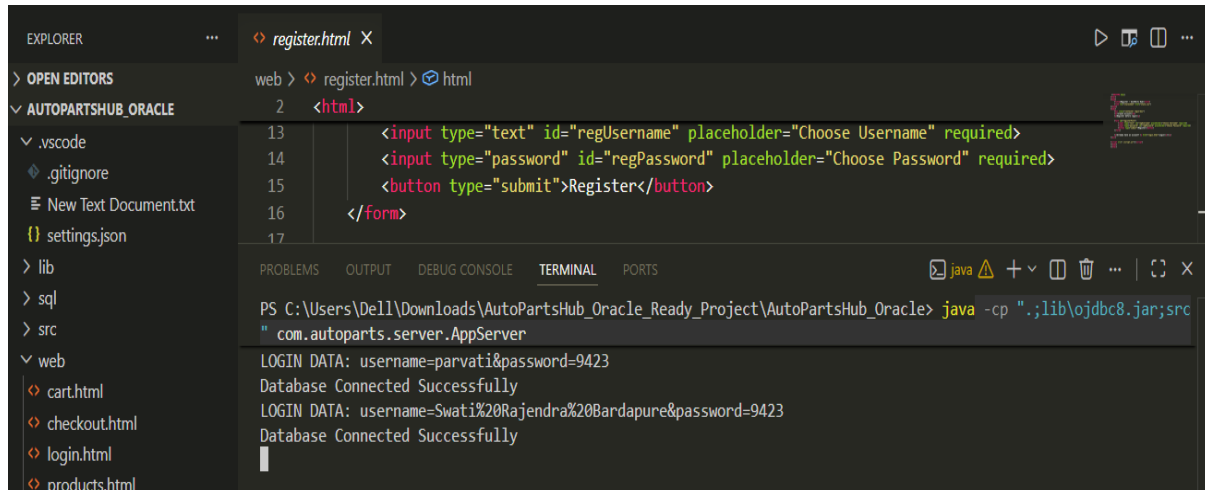


Caption: Cart Functionality

5. Database



6. Backend Running



The screenshot shows a VS Code editor with a file explorer on the left. The file explorer shows a project structure with folders like .vscode, .gitignore, New Text Document.txt, settings.json, lib, sql, src, and web. The web folder is expanded, showing files like cart.html, checkout.html, login.html, and products.html. The register.html file is open in the editor, showing HTML code for a registration form. The terminal window at the bottom shows the command to run the Java application and the output of the application, which includes login data and database connection status.

```
register.html
2 <html>
13 <input type="text" id="regUsername" placeholder="Choose Username" required>
14 <input type="password" id="regPassword" placeholder="Choose Password" required>
15 <button type="submit">Register</button>
16 </form>
17

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Dell\Downloads\AutoPartsHub_Oracle_Ready_Project\AutoPartsHub_Oracle> java -cp ".;lib\ojdbc8.jar;src"
"com.autoparts.server.AppServer"
LOGIN DATA: username=parvati&password=9423
Database Connected Successfully
LOGIN DATA: username=Swati%20Rajendra%20Bardapure&password=9423
Database Connected Successfully
```

PERFORMANCE TESTING Test Cases:

Register new user → Success

Register duplicate → Error

Login correct → Success

Login incorrect → Fail

Add to cart → Works

Place order → Success

Result.

Performance testing was conducted to evaluate the efficiency and correctness of the AutoParts Hub system under various user scenarios. Different test cases were executed to verify the functionality of each module. The system was tested for user registration by creating a new user account, which was successfully stored in the database.

An attempt to register a duplicate username resulted in an appropriate error, ensuring data integrity. The login functionality was also tested with both valid and invalid credentials. Valid login details allowed access to the system, while incorrect credentials displayed an error message.

Further testing included the add-to-cart functionality, which worked correctly by updating the cart count and total amount using local storage. The order placement feature was also verified, and orders were successfully processed and confirmed.

Result: The system performed efficiently and produced correct results under all tested conditions, demonstrating reliability and proper functionality.

CHALLENGES FACED

During the development of the AutoParts Hub project, several technical challenges were encountered. One of the major issues was the “port already in use” error, which prevented the backend server from starting properly. Additionally, there were instances where the backend server was not responding, causing failures in frontend-backend communication.

Another challenge involved handling Fetch API errors, where requests from the frontend failed due to incorrect endpoints or server issues. The duplicate username constraint in the database also caused errors during registration when the same username was entered multiple times.

SOLUTIONS IMPLEMENTED

To resolve the issues faced during development, appropriate solutions were implemented. The port conflict problem was addressed using command-line tools such as netstat and taskkill to identify and terminate processes using the same port. Backend connectivity issues were resolved by ensuring that the server was running correctly and that the API endpoints matched the frontend requests. Fetch API errors were fixed by correcting URLs and handling exceptions properly in the Java backend.

The duplicate username issue was managed by implementing exception handling in the database layer, which prevented duplicate entries and displayed meaningful error messages. Additionally, users were guided to use unique usernames during registration.

FUTURE SCOPE

The AutoParts Hub system developed in this project provides a basic yet functional e-commerce platform. However, there is significant scope for further enhancement to make the system more advanced, secure, and user-friendly.

One of the major future improvements includes the integration of a payment gateway, which will allow users to make secure online payments using debit cards, credit cards, or digital wallets. This will transform the application into a complete e-commerce solution.

An admin dashboard can also be implemented to enable administrators to manage products, users, and orders efficiently. Through this dashboard, admins can add, update, or delete products and monitor system activity. The system can be enhanced by adding product images, descriptions, and user ratings, which will improve user experience and help customers make better purchasing decisions. Another important improvement is the development of a responsive mobile-friendly design, ensuring that the application works smoothly across different devices such as smartphones and tablets.

REFERENCES

Oracle Database Documentation

Java Documentation

GitHub

W3Schools

Stack Overflow

CONCLUSION

The AutoParts Hub project successfully demonstrates the design and development of a full-stack e-commerce web application using modern technologies. The system integrates a user-friendly frontend developed with HTML, CSS, and JavaScript, a Java-based backend using HttpServer, and an Oracle database connected through JDBC.

The application fulfills its primary objectives by allowing users to register, log in, browse products, add items to a cart, and place orders. All core functionalities were tested and executed successfully, ensuring that the system operates reliably under different conditions. This project provided valuable hands-on experience in full-stack development, database management, and client-server architecture. It also helped in understanding the practical implementation of theoretical concepts such as HTTP communication, data handling, and user authentication.

In conclusion, the AutoParts Hub system serves as a strong foundation for building scalable and secure e-commerce applications. With further enhancements such as payment integration, improved security, and advanced user features, the system can be extended into a fully functional commercial platform.

GITHUB REPOSITORY

The complete source code of the AutoParts Hub project is available on GitHub for reference and evaluation.

Project Link:

<https://github.com/swatibardapure9423-commits/upskillcampus>